

PRELIMINARY INVESTIGATIONS OF ANTIMICROBIAL SCREENING OF CRUDE EXTRACTS OF SPONGES AND GORGONIANS SPECIES FROM SAUDI RED SEA COAST

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ABSTRACT

Some of the marine organisms have been found to possess chemotherapeutic properties. The active ingredients can be used as antibacterial, antifungal, antitumor and antiviral agents. These compounds are sometimes difficult to be synthesized in laboratory. Since very little work has been done in the past on the organisms found in the Red Sea, therefore some sponges and gorgonians were collected from the sea near the coast of Jeddah. Our results indicated that *Sh. boydii*, *E. coli*, and *S. aureus*, exhibited susceptibility to some crude extracts of sponges and gorgonians. Ampicillin, Amoxicillin and Cefuroxime were used as standard drugs. During the screening of antifungal activity, it was noted that *Elisella sp.* and *Axinellida sp.* showed significant activity against *C. lunata*, *T. longifusus* and *M. canis*. Myconazole, and Ketoconazole were used as standard drugs for comparison.

INTRODUCTION

The history of drugs is woven with plants, animals and minerals from earlier times. They are mostly of terrestrial origin with smaller number coming from the sea. Despite the wide availability of medicinally useful compounds a continuing search for new antimicrobial agents is still the primary need of the time. These compounds are generally smaller molecules ideally suited to serve either themselves or through chemical modifications as potential new pharmaceutical agents for the treatment of variety of ailments. The scientific community is focusing its efforts on the isolation and characterization of biologically active compounds derived from marine organisms. It has been demonstrated that marine

organisms are excellent sources of new drugs, (Faulkner, 1995; Fenical, 1996). One may find exotic compounds arranged in completely different ways from those found on land plants and animals. Some of the antibiotics may have drawbacks in the sense that they have either limited antimicrobial spectrum or are associated with some side effects or contraindications. The combination of the genetic versatility of the microbes and widespread over use of antibiotics has also led to increasing clinical resistance of previously sensitive microorganisms and the emergency of previously uncommon infections. It has therefore become highly desirable to explore new molecules exhibiting prominent activity against pathogenic organisms with less undesirable effects (Vlietinck, 1987). In the early 1950's, the first

Table Ia
Antibacterial Activity of Crude Extracts of sponges

Bacterial cultures	Zone of inhibition in mm									Standard drugs		
	<i>Axinellide sp.</i>	<i>Phyllospongia sp.</i>	<i>Acanthella carteri</i>	<i>Gravella cyatophora</i>	<i>Hyrtios erecta</i>	<i>Subarea mollis</i>	<i>Siphonochalina sp.</i>	<i>Ircinia sp.</i>	<i>Sinularia sp.</i>	Ampicillin	Amoxicillin	Cefuroxime
<i>S. aureus</i>	8	7	7	7	7	7	7	8	8	22	21	21
<i>S. pyogenes</i>	6	-	-	-	-	-	-	6	6	20	20	20
<i>A. hydrophilla</i>	6	7	-	-	-	7	-	6	7	19	19	19
<i>C. diphtheriae</i>	-	7	6	-	-	6	-	8	-	16	-	15
<i>B. subtilis</i>	-	9	-	-	-	-	-	8	-	18	19	19
<i>E. soli</i>	6	6	6	6	6	6	6	6	6	-	-	19
<i>S. typhi</i>	-	8	-	-	-	8	-	9	-	21	20	18
<i>Sh. Boydii</i>	6	6	6	6	-	7	6	6	6	21	21	18
<i>K. pneumoniae</i>	-	-	-	-	-	-	-	-	-	9	-	19
<i>P. mirabilis</i>	-	7	-	-	-	-	-	7	6	20	20	20
<i>Ps. Aeruginosa</i>	-	-	-	-	-	-	-	6	6	12	-	-

Key: mm: Zone of Inhibition. - No Antibacterial Activity. Concentration of test sample: 200 µg/100 µl.

antiviral agent was isolated from a sponge *Cryothya crypta* (Bergmann and Burke 1955), two of the unusual nucleosides (spongouridine and spongoadenosine), proved to have antiviral activity. The discovery subsequently led to a series of analogous compounds synthesized in the laboratory to be used as chemotherapeutic agents. The diversity of sponge species is a significant feature of marine organisms, in the biological

community of the Red Sea. In view of the necessity of developing effective drugs to combat modern health crisis, sponges are becoming increasingly important as a source of bioactive substances.

Compounds extracted from sponges are considered as drug of choice as they have less side effects or after effects as compared to the drugs synthesized in the laboratory. Such

Table 1b:
Antibacterial activity of Crude Extracts of Gorgonians

Bacterial cultures	Zone of Inhibition in mm			Standard Drugs (200 µg/100µl)		
	<i>Rumphella aggregata</i>	<i>Elisella sp.</i>	<i>Acarboria erythraea</i>	Ampicillin	Amoxicillin	Cefuroxim
<i>S. aureus</i>	-	7	-	22	1	21
<i>S. pyogenes</i>	-	-	-	20	20	20
<i>A. hydrophilla</i>	-	-	-	19	19	19
<i>C. diphtheriae</i>	6	-	6	16	-	15
<i>B. subtilis</i>	-	-	-	18	19	19
<i>E. coli</i>	6	6	6	-	-	19
<i>S. typhi</i>	-	-	-	21	20	18
<i>Sh. boydii</i>	6	-	6	21	21	18
<i>K. pneumoniae</i>	-	-	-	9	-	19
<i>P. mirabilis</i>	-	-	-	20	20	20
<i>Ps. aeruginosa</i>	-	-	-	12	-	-

Key: mm: Zone of Inhibition, - No Antibacterial Activity. Concentration of test sample: 200 µg/100µl

Table 2
MIC Values of Crude Extracts of the Sponges against Pathogenic Bacteria

Bacterial Culture	MIC Value in µg/ml				
	<i>Ircinia sp.</i>	<i>Phyllospangia sp.</i>	<i>Suberea mollis</i>	<i>Axinellida sp.</i>	<i>Sinularia sp.</i>
<i>Staphylococcus aureus</i>	90	-	-	90	95
<i>Cornebacterium diphtheriae</i>	85	-	-	-	-
<i>Bacillus subtilis</i>	85	80	-	-	-
<i>Salmonella typhi</i>	75	85	90	-	-

compounds exhibit a diversity of chemical structure (Scheuer, 1991 and 1995). Some of the sponges and related organisms have been found to be associated with antibacterial, antifungal, antiviral and antitumor activity (Higa, 1994, Avasti and

Bhakumi, 1993). The present work describes the collection and identification of various species of sponges and gorgonians and effects of their crude extract on some gram positive and gram-negative bacteria and some of the fungi.

Table 3
Antifungal activity of crude extracts of Sponges and Gorgonians

Fungal Culture	Zone of Inhibition in mm							Standard Drugs (200 ttg/100 pfl)		
	<i>Suberera mollis</i>	<i>Axinellida sp.</i>	<i>Biemna sp.</i>	<i>Simularia sp.</i>	<i>Siphonochalina sp.</i>	<i>Acanthella carteri</i>	<i>Phyllospongia sp.</i>	<i>Hyrtios erecta</i>	Myconazole	ketoconazole
<i>Trichophyton senai</i>	+	+++	+	+	++	-	+	+	+++	+++
<i>Candida albicans</i>	-	+	++	-	-	-	-	-	+++	+++
<i>Trichophyton</i>	+	++	+	+	++	-	+	-	+++	+++
<i>Allescheria boydii</i>	-	++	+	-	+	-	-	+	+++	+++
<i>Microsporuni canis</i>	++	++	++	-	+++	++	+++	++	+++	+++
<i>Aspergillus niger</i>	+	+	+	+	+	+	+	+	+++	+++
<i>Drechslera rostrata</i>	+	+	++	-	+	++	+++	++	+++	+++
<i>Nigrospora oryzae</i>	-	+	+	+	+	+	+	-	+++	+++
<i>Stachyotrys atm</i>	++	++	++	++	++	++	++	++	+++	+++
<i>Otrydaria lunata</i>	+++	+++	+++	+++	+++	+++	+++	+++	+++	+++
<i>Trichophyton longillius</i>	++	+++	++	+++	++	++	+++	+++	+++	+++

Key: (+++) = Strong inhibition, (++) = Medium inhibition, (+) = Weak inhibition and (-) No activity.
Concentration of test sample: 400 µg/ml of media.

MATERIALS AND METHODS

Sponges and gorgonians were collected from the coastal area of Saudi Red Sea, near the coast of Jeddah at a depth of approximately 18-25 meters (Figure 1). The organisms were then washed with sea-water, and the extraneous material was removed. They were carefully cleaned and cut into small pieces. Later they were ground in blender and steeped first with methanol and second time with chloroform, methanol (2:1, v/v); each time at room temperature for 24 hours. The solvent was removed under vacuum at a temperature of less than 40° C. The two extracts were combined, and the resultant concentrated matter was finally evaporated to form a crude extract by evaporation under nitrogen.

The antimicrobial activity was determined by Agar well diffusion method (Akaniro *et al*, 1990). One loopful of 24

hours old bacterial culture containing approximately 10^4 - 10^6 cfu (colony forming units) was spread on the surface of Mueller Hinton Agar (MHA) plates, wells were dugged in the medium with the help of sterile metallic borer. Stock solutions of the extracts were prepared and different dilutions were added in their respective wells. The extent of activity of the crude extracts were compared to standard antibacterial agents of known activity such as ampicillin, amoxicillin and cefuroxime under the same test conditions. All the experimental plates were incubated at 37°C for 18-24 hours and activity was determined by measuring the diameter of zones showing complete inhibition of growth (in mm). The Minimum Inhibitory Concentration (MIC) is defined as the lowest concentration, which prevent the growth of test organisms (bacteria and fungi etc.) under optimal conditions. The method used to determine MIC is serial dilution in nutrient broth, which is based on the incubation of the

Table 4
MIC Values of crude extracts of the sponges and Gorgonians against pathogenic Fungi

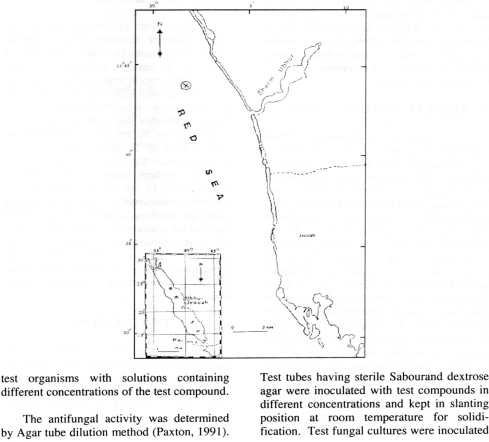
Fungal culture	MIC Values in µg/ml						
	<i>Acanthella carteri</i>	<i>Hyrtios erecta</i>	<i>Suberera mollis</i>	<i>Siphonochalina sp.</i>	<i>Sinularia sp.</i>	<i>Rumphella aggregata</i>	<i>Axinellida sp.</i>
<i>Microsporum canis</i>	-	-	-	275	-	200	200
<i>Trichophyton longifusus</i>	-	250	-	-	325	250	225
<i>Curvuluriu lunata</i>	350	225	250	300	300	275	250

Unit = µg/ml.

Table 4
MIC Values of crude extracts of the sponges and Gorgonians against pathogenic Fungi

Fungal culture	MIC Values in µg/ml						
	<i>Acanthella carteri</i>	<i>Hyrtios erecta</i>	<i>Suberera mollis</i>	<i>Siphonochalina sp.</i>	<i>Sinularia sp.</i>	<i>Rumphella aggregata</i>	<i>Axinellida sp.</i>
<i>Microsporum canis</i>	-	-	-	275	-	200	200
<i>Trichophyton longifusus</i>	-	250	-	-	325	250	225
<i>Curvalaria lunata</i>	350	225	250	300	300	275	250

Unit = µg/ml.



test organisms with solutions containing different concentrations of the test compound.

The antifungal activity was determined by Agar tube dilution method (Paxton, 1991).

Test tubes having sterile Sabourand dextrose agar were inoculated with test compounds in different concentrations and kept in slanting position at room temperature for solidification. Test fungal cultures were inoculated

on the slant and growth inhibitions were observed after incubation period of 7 days (Washington and Sutter, 1980). Myconazole and Ketoconazole were used as standard antifungal drugs. MIC values of the extracts were determined by the same procedure.

In order to evaluate the antimicrobial activity among the gram positive groups, the bacteria used were *Staphylococcus aureus*, *Streptococcus pyogenes*, *Aeromonas hydrophilla*, *Cornebacterium diphtheriae*, and *Bacillus subtilis* and among the gram negative *Escherichia coli*, *Salmonella typhi*, *Shigella boydii*, *Klebsiella pneumonia*, *Proteus mirabilis*, *Pseudomonas aeruginosa*.

For the antifungal activity, the organisms used were *Trichophyton mentagrophyte*, *Allerscheria boydii*, *Microsporum canis*, *Aspergillus niger*, *Drechslera rostrata*, *Negrospora oryzae*, *Stachybotrys atra*, *Curvularia lunata*, *Trichophyton semii* and *Trichophyton longifusus*.

RESULTS AND DISCUSSION

Marine sponges use diversity of unusual compounds, they develop these chemicals a self defence mechanism against predators, inhibiting pathogenic microbes, discouraging the formation of a bacterial biofilm and subsequently microfouling community. Nine Sponges and three gorgonians extracts were examined. The antibacterial activity is summarized in Table. 1 (a and b), *S. aureus*, *S. pyogenes*, *S. typhi*, *Shigella boydii*, *B. subtilis*, and *C. diphtheriae* exhibited susceptibility against *Ircinia sp.*, MIC of significantly active samples were determined using broth dilution method and values are shown in Table-2. Ampicillin, amoxicillin and cefuroxime were also tried to compare the extent of activity. The antifungal activity of crude extracts of sponges and gorgonians are given in Table-3 using a battery of pathogenic fungi. These are *Trichophyton mentagrophyte*, *Allescheria boydii*, *Microsporum canis*, *Aspergillus niger*,

Drechslera rostrata, *Nigrospora oryzae*, *Stachybotrys atra*, *Curvularia lunata*, *Trichophyton semii*, and *Trichophlongifusus*. It was observed that *Axinellida sp.* and *Phyllospongin* exhibited significant antifungal activity against *M. canis*, *T. longifusus* and *C. lunata*. These were almost as active as standard drugs i.e. Myconazole and Ketoconazole; *C. lunata*, and *T. longifusus*. were found to be susceptible to the crude extracts of sponges like *Acanthella carteri*, *Subera mollis* and *Siphonochalina sp.* MIC values of active samples are given in Table 4. Although the pattern of antimicrobial activity was different in different species, but in general the findings are in accordance with the work carried out by other groups in different parts of the world.

Newbold et al. (1999) reported that the crude extract of *Axinella* species exhibited antimicrobial activity and isolated a compound, stevensine; responsible for this activity. The present study in our laboratory indicated *Siphonochaline sp.* to be effective against *S. aureus*, *B. subtilis* and *Sh. boydii*, a similar observation was made by Ajabnoor et al.(1991) when he collected and tested the crude extracts of 22 sponges collected from the Red Sea near the coast of Jeddah. Another group of workers, Faulkner et al. (1985), observed that 11 out of 40 sponge species exhibited antimicrodial activity. A French group of workers. (Amade et al. 1982) collected the sponges from the Brittany coast and when subjected to bioassay screening, promising activity was shown by *Homaxinella*, *Hyrtios*, *Verongia* and *Dysdiae* species, *Siphonochalina* against Gram positive bacteria.

CONCLUSIONS

Sponges that are apparently of no commercial value, today are rich sources of bioactive substances. They contain a diversity of unusual compounds. Bioassay of the extract from a variety of species of sponges in the Red Sea have indicated the presence of a

diversity of substances whose structure when isolated may not have terrestrial counterpart. The pattern of the antimicrobial activity varied with species. However, it can be concluded that *Ircinia* sp., *Simularia* sp. and *Phyllospongia* sp. exhibited more promise for future studies. Some of these extracts have shown definite activity as antifungal agents and may prove to be potentially important and may be developed into value added products by preparing their derivatives.

An overall picture of these tables unveils the fact that these crude extracts were more active as antifungal agent rather than antibacterial agent. Moreover the extract may contain more than a dozen compounds in all. If the active principle is isolated in a pure form then one might get better results; it may also act as a starting material for semi-synthetic drugs, and it may prove to efficacious in medicinal chemistry.

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REFERENCES

- Akaniro, J.C. Vidaurre, C.E., Stuttman, H.R., and Marks, M.I., (1990). Antimicrobial agents and chemotherapy; American Society of Microbiology, Washington, U.S.A., 34: 1880-1884.
- Amade, P., Psedano, D., and Chevolot, L., (1982). Antimicrobial activity of marine Sponges, from French Polynesia and Brittany. *Mar. Biol.* 70: 223-228.
- Avasthi, K. and Bhakuni, D.S., (1993). Marine Nucleosides. *Indian J. Of Het. Chem.* 2: 203-218.
- Bergmann, W., and Burke, D.C. (1955). The nucleoside of sponges, spongothy-midine and spongouridine. *J. Org. Chem.* 20: 1501-1507.
- Faulkner, D.J., Thompson, J.E. and Walker, R.P., (1985). Screening and bioassay for biological active substances from forty marine sponges from San Diego, California. *Mar. Biol.* 88: 11-12.
- Faulkner, D.J., (1995). Chemical Riches from the Oceans. *Chem. In Britain*, 31: 680684.
- Fenicle, W. (1996). Status of new drugs from marine organisms. *Oceanography* 9(1): 23-27.
- Higa, T., Tanaka, T.I., Kitamura, A., Kyoma, T., Takashi, M., and Uchida, T., (1993). Bioactive compounds from marine sponges. *Pure and App. Chem.* 66: 2227-2230.
- Newbold, R.W., Jensen, P.R., Fenical, W. and Pawlik, J.R. (1999). Antimicrobial activity of Caribbean Sponge extracts. *Aq. Microb. Ecol.* 19: 279-284.
- Numata, A. and Iritani, M. (1997). Novel antitumor metabolites produced from a fungal strain from a sea hare. *Tetrahedron. Lett.* 88(47): 8215-8218.
- Paxton, I.D., (1991). Methods in Plant Biochemistry, Assays for antifungal activities, Academic Press, London. 3rd. Ed., 6: pp. 33-45.
- Scheuer, P.J. (1991). Drugs from the Sea. *Chem. and Ind.* 5: 276-279.
- Scheuer, P.J. (1995). Marine Natural Products Research; A Look into the Dive Bag. *J. of Nat. Prod.* 58: 335-343.
- Tringali, C., (1997). Bioactive metabolites from marine algae, recent results, Benthan Science Publishers, London, Vol. I, pp. 375-394.
- Viletinck, A.J., (1987). Topics in Pharmaceutical Sciences, Elsevier Science Publication, Amsterdam, pp. 249-262.
- Washington, J.A., and Sutter, V.L., (1980). Agar and Microbroth dilution procedures, American Society of Microbiology, Washington, 3rd. Ed., pp. 453-462.